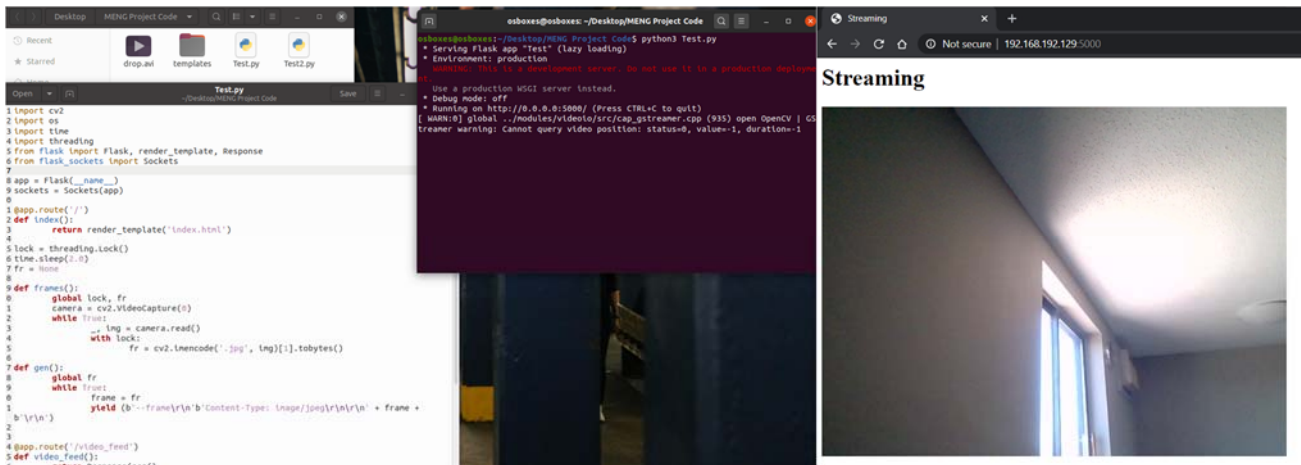


Project Update – Deep Singh

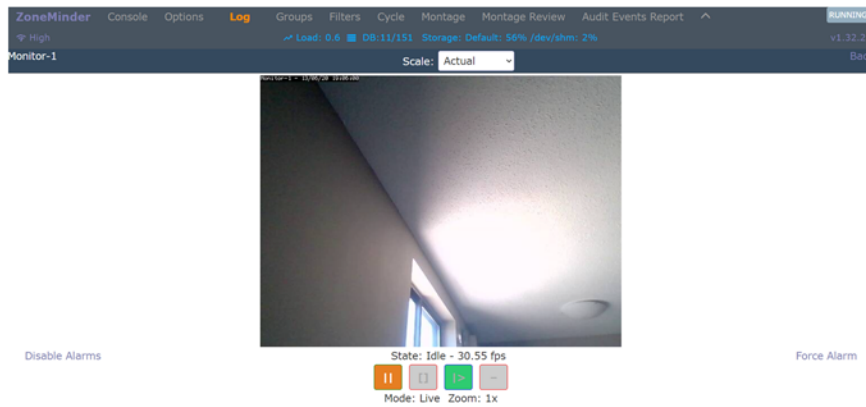
1. Deployed Linux VMs for **Monitoring Servers** and **Raspbian VM** to emulate remote device.
2. Tested **MotionEye**, **Zoneminder** and **iSPY**. Found ZOneminder to best suited for our project.
3. Tested **API** and **Websockets** for Webstreaming
4. Websockets has better performance so **Developed Websocket code** with **multithreading** to support multiple clients. **(Figure 1)**
5. Successfully deployed Zoneminder and added IP camera. **(Figure 2 and 3)**
6. Developed Script to switch Pedestrian detection from server (In case of RSU failure)
7. Tested **OpenWisp** and **Nagios** for RSU management. Found Nagios has better features for Management.
8. Deployed Server for Monitoring and health check of multiple RSU – Using Nagios (I'm using XI, but we can also deploy same script on Nagios core) **(Figure 4,5 and 6)**
9. Included system has provision for **email and mobile** notifications also **(Figure 7)**



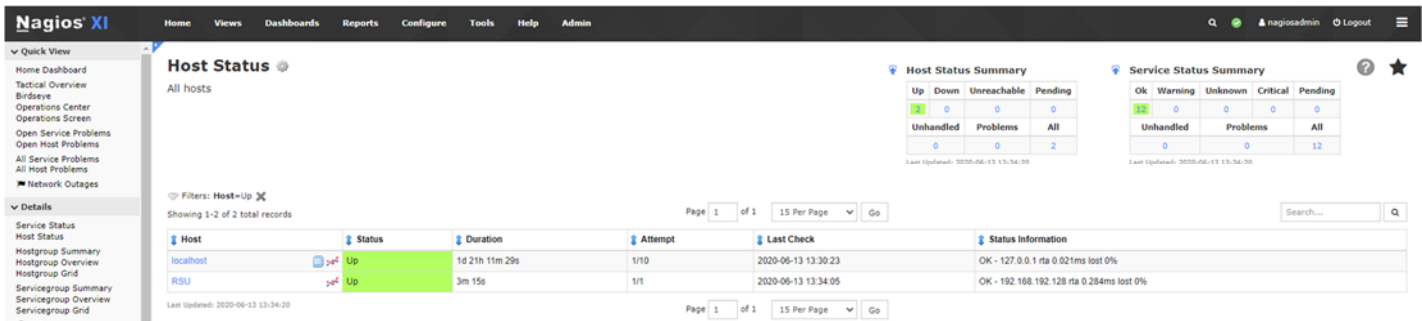
(Figure 1)

ZoneMinder Console Options Log Groups Filters Cycle Montage Montage Review Audit Events Report										
v1.32.2										
Name: text or regular expression Function: All Status: All Source: text or regular expression Monitor: All										
Capturing 100%										
ADD EDIT DELETE SELECT										
Name	Function	Source	Total	Hour	Day	Week	Month	Archived	Zones	All
Monitor-1	Monitor Capturing 0.00 fps 0B/s	192.168.192.129:5000	0 0B	0B	0B	0B	0B	0 0B	1	☐ ☒
Pedestrian Detection feed	Monitor Capturing 25.00 fps 1.52MB/s	192.168.192.128:5000	2 197.33kB	2 197.33kB	2 197.33kB	2 197.33kB	2 197.33kB	0B	1	☐ ☒
Ip Webcam	Mocord Capturing 25.00/25.00 fps 864kB/s	192.168.0.22:8080	3 765.22kB	3 765.22kB	3 765.22kB	3 765.22kB	3 765.22kB	0B	1	☐ ☒
	2.36MB/s		5 0.94MB	5 0.94MB	5 0.94MB	5 0.94MB	5 0.94MB	0 0B	3	

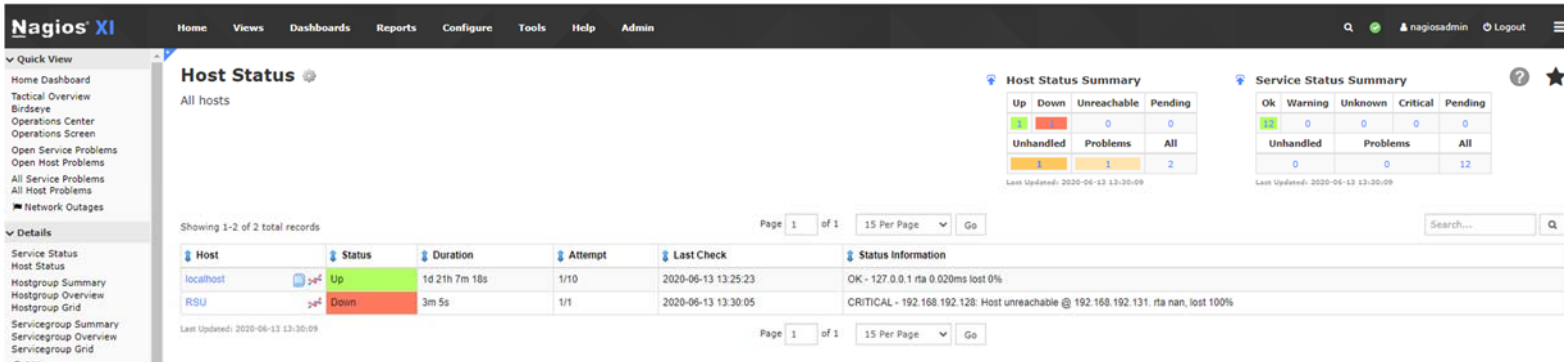
(Figure 2)



(Figure 3)



(Figure 4)

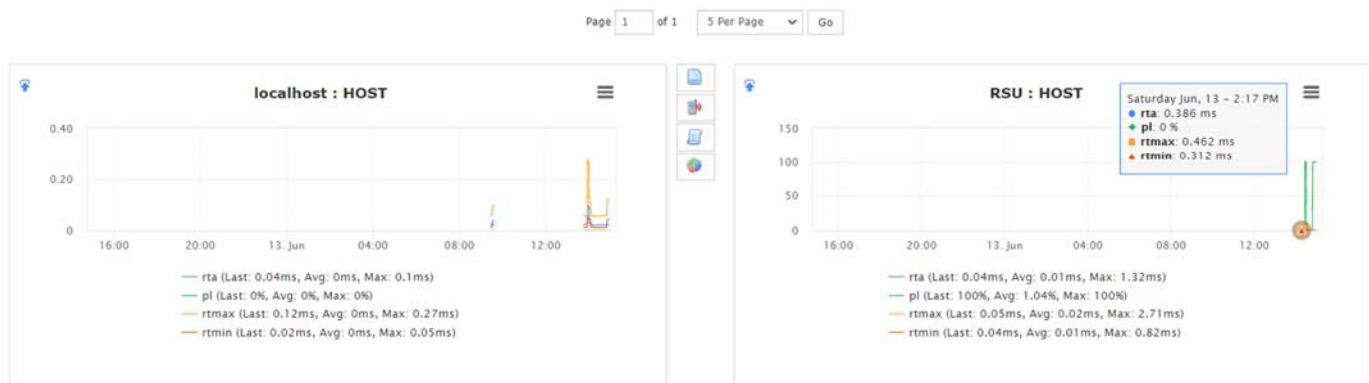


(Figure 5)

Performance Graphs

Host Performance Graphs - 24 Hour View

Showing 1-2 of 2 total records



(Figure 6)

Email Settings

Modify the settings used by your Nagios XI system for sending email alerts and informational messages.

[Send a Test Email](#)

[Outbound](#) [Inbound](#)

Note: Mail messages may fail to be delivered if your XI server does not have a valid DNS name. For more information, read [Understanding Email Sending in Nagios XI](#).

Outbound Mail Settings

Send From:

Send Method: ☐ Sendmail
☒ SMTP

Logging: ☒ Enable logging of mail sent with the internal mail component (PHPMailer) /usr/local/nagiosxi/tmp/phpmailer.log

SMTP Settings

Host: ?

Port:

Username:

Password: [Show/Hide](#)

Security: ☐ None
☒ TLS
☐ SSL

[Send a Test Email](#)

[Update Settings](#) [Cancel](#)

(Figure 7)